

EE596H: Quiz

Total Questions: 2, Total Marks: 30, Time: 60 Mins

1. Consider the following function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} \mathbf{x} + \mathbf{x}^T \begin{bmatrix} 3 \\ 5 \end{bmatrix} + 6$$

- (a) Find the gradient and Hessian of f at the point $[1, 1]^T$. 5
- (b) Find the directional derivative of f at $[1, 1]^T$ with respect to a unit vector in the direction of maximal rate of increase. 5
- (c) Find a point that satisfies the FONC (interior case) for f . Does this point satisfy the SONC (for a minimizer)? 5

Total: 15

2. For each of the following matrices determine whether they are positive/negative semidefinite/definite or indefinite:

(a)

$$\mathbf{A} = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{bmatrix}$$

5

(b)

$$\mathbf{B} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

5

(c)

$$\mathbf{C} = \begin{bmatrix} -5 & 1 & 1 \\ 1 & -7 & 1 \\ 1 & 1 & -5 \end{bmatrix}$$

Total: 15